TITLE:

DIGITAL DISTRACTION ANALYSER (CLI)

Name: Chinmay Jain

Registration Number: 25BCE10776

Branch: B. Tech Computer Science and Engineering

Year: 1st Year

Course Code: CSA2001

Course Title: Fundamentals in AI and ML

Date: 31st March 2026

INTRODUCTION:-

In today's digital world, students spend a lot of time on mobile phones, social media, and gaming and many more. This often leads to reduced productivity and poor time management.

This project aims to analyse a user's daily digital usage and provide useful suggestions to improve their productivity. It is a simple CLI-based system that interacts with the user and gives feedback based on their habits.

MOTIVATION OF THE PROJECT:-

- The main motivation behind this project is to help students become more aware of their screen time and digital habits.
- Many students waste time unknowingly on social media and entertainment. This system helps them understand their usage and encourages them to manage their time better.

PROBLEM STATEMENT:-

Students often spend excessive time on digital platforms without realizing its impact on productivity.

The problem is to create a system that can analyse daily digital usage and provide meaningful suggestions to improve productivity.

OBJECTIVE OF THE PROJECT:-

1) To analyse user's daily digital usage.
2) To classify productivity level (Low, Medium, High).
3) To provide useful suggestions based on usage.
4) To create a simple and interactive CLI-based system.

EXISTING METHODS:-

Currently, users manually track their screen time using mobile apps or settings.
However, these methods only show usage data and do not provide meaningful suggestions or analysis.

PROS AND CONS OF EXISTING METHODS

Pros:
- Easy to access screen time data.
- Built-in mobile features available.

Cons:
- No proper analysis.
- No personalized suggestions.
- Users often ignore the data.

HARDWARE AND SOFTWARE REQUIREMENTS:-

Hardware:

- Laptop or computer.

Software:

- Python.
- Command Prompt / Terminal.

METHODOLOGY AND GOAL:-

- The system takes user input for daily digital usage such as YouTube, social media, and gaming hours.
- It then calculates total distraction time and uses rule-based logic (if-else conditions) to classify productivity levels.
- Based on this classification, the system provides suggestions to improve habits.

FUNCTIONAL MODULES DESIGN AND ANALYSIS:-

1) Input Module: Takes user input (screen time hours).
2) Validation Module: Ensures only valid numeric input is accepted.
3) Processing Module: Calculates total distraction.
4) Decision Module: Determines productivity level.
5) Output Module: Displays suggestions.

6) Feedback Module: Takes user feedback.
7) Loop Module: Allows repeated usage.

ALGORITHM DEVELOPMENT:-

1) Start the program
2) Take user name
3) Take input for YouTube, social media, gaming, and sleep
4) Validate inputs (only numbers allowed)
5) Calculate total digital distraction
6) Apply conditions:

- If distraction ≥ 8 → Low productivity
- If distraction ≥ 4 → Medium productivity
- Else → High productivity

7) Display suggestions
8) Take feedback (yes/no)
9) Ask user to repeat or exit
10) End program

CODING:-

```
def get_valid_input(message):
    attempts = 0
    while True:
        if attempts >= 3:
            print("\nToo many invalid attempts. Exiting
program.")
```

```
        exit()

    value = input(message)

    if value.isdigit():
        return int(value)
    else:
        print("Invalid input, please enter numbers
only.")
        attempts += 1


# Main Program
name = input("Enter your name: ")

while True:
    print("\nHii,", name)
    print("Let's analyze your daily digital usage\n")

    # Taking inputs one by one
    yt = get_valid_input("Enter YouTube hours per
day: ")
    insta = get_valid_input("Enter Instagram/Social
Media hours per day: ")
    game = get_valid_input("Enter Gaming hours
per day: ")
    sleep = get_valid_input("Enter Sleep hours: ")

    distraction = yt + insta + game
```

```
    print("\n--- Analysis Report ---")
    print("Total Digital Distraction:", distraction,
"hours")

    if distraction >= 8:
        print("Productivity Level: Low")
        print("Suggestions:")
        print("• Limit social media usage to 1-2 hours
per day")
        print("• Set a daily study/work schedule")
        print("• Keep your phone away while
studying")
        print("• Try work for 25-minutes then 5-
minutes break (Pomodoro technique)")
    elif distraction >= 4:
        print("Productivity Level: Medium")
        print("Suggestions:")
        print("• Reduce unnecessary screen time
gradually")
        print("• Plan your day with clear goals")
        print("• Take short breaks instead of long
scrolling sessions")
        print("• Focus on completing important tasks
first")
    else:
        print("Productivity Level: High")
        print("Suggestions:")
        print("• Maintain your current routine")
```

```
    print("• Keep tracking your screen time
regularly")
    print("• Try learning new skills in free time")
    print("• Stay consistent and avoid
distractions")

  # Feedback section
  while True:
    feedback = input("\nWas this analysis
helpful? (yes/no): ").lower()

    if feedback == "yes":
      print("I'm glad you liked it!")
      break
    elif feedback == "no":
      print("Thanks for your feedback, we will
improve!")
      break
    else:
      print("Invalid input, please enter 'yes' or
'no'.")

  # Loop for reuse
  while True:
    again = input("\nDo you want to analyze
again? (yes/no): ").lower()

    if again == "yes":
      break
```

```
    elif again == "no":
        print("\nThank you for using the system!")
        exit()
    else:
        print("Invalid input, please enter 'yes' or
'no'.")
```

OUTPUT:-

1.Program Start Screen:-

```
= RESTART: C:\Users\CHINMAY JAIN\AppData\Local\Programs\Python\Python313\distraction.py
Enter your name: chinmay

Hii, chinmay
Let's analyze your daily digital usage

Enter YouTube hours per day:
Enter Instagram/Social Media hours per day:
Enter Gaming hours per day:
Enter Sleep hours:
```

2.Invalid Attempt Limit:-

```
Hii, chinmay
Let's analyze your daily digital usage

Enter YouTube hours per day: a
Invalid input, please enter numbers only.
Enter YouTube hours per day: g
Invalid input, please enter numbers only.
Enter YouTube hours per day: d
Invalid input, please enter numbers only.

Too many invalid attempts. Exiting program.
```

3.Invalid Input Handling:-

```
Hii, chinmay
Let's analyze your daily digital usage

Enter YouTube hours per day: a
Invalid input, please enter numbers only.
Enter YouTube hours per day:
```

4.Output Low Productivity:-

```
= RESTART: C:\Users\CHINMAY JAIN\AppData\Local\Programs\Python\Python313\distraction.py
Enter your name: chinmay

Hii, chinmay
Let's analyze your daily digital usage

Enter YouTube hours per day: 7
Enter Instagram/Social Media hours per day: 5
Enter Gaming hours per day: 1
Enter Sleep hours: 5

--- Analysis Report ---
Total Digital Distraction: 13 hours
Productivity Level: Low
Suggestions:
• Limit social media usage to 1-2 hours per day
• Set a daily study/work schedule
• Keep your phone away while studying
• Try work for 25-minutes then 5-minutes break (Pomodoro technique)
```

5.Output Medium Productivity:-

```
Hii, chinmay
Let's analyze your daily digital usage

Enter YouTube hours per day: 2
Enter Instagram/Social Media hours per day: 3
Enter Gaming hours per day: 1
Enter Sleep hours: 6

--- Analysis Report ---
Total Digital Distraction: 6 hours
Productivity Level: Medium
Suggestions:
• Reduce unnecessary screen time gradually
• Plan your day with clear goals
• Take short breaks instead of long scrolling sessions
• Focus on completing important tasks first
```

6.Output High Productivity:-

```
Hii, chinmay
Let's analyze your daily digital usage

Enter YouTube hours per day: 1
Enter Instagram/Social Media hours per day: 1
Enter Gaming hours per day: 1
Enter Sleep hours: 8

--- Analysis Report ---
Total Digital Distraction: 3 hours
Productivity Level: High
Suggestions:
• Maintain your current routine
• Keep tracking your screen time regularly
• Try learning new skills in free time
• Stay consistent and avoid distractions
```

7.Output Feedback Yes:-

```
Was this analysis helpful? (yes/no): yes
I'm glad you liked it!
```

8.Output Feedback No:-

```
Was this analysis helpful? (yes/no): no
Thanks for your feedback, we will improve!
```

KEY IMPLEMENTATIONS OUTLINE OF THE SYSTEM:-

- Rule-based decision making.
- Input validation with attempt limit.
- Feedback system.
- Loop for repeated analysis.

SIGNIFICANT PROJECT OUTCOMES:-

- It Helps users understand their digital habits.
- Encourages better time management.

- Improves productivity awareness.

TESTING AND REFINEMENT:-

1) The system was tested with:
2) Valid inputs
3) Invalid inputs (letters, wrong values)
4) Attempt limit conditions
5) Feedback system
6) All features were refined to ensure proper functioning.

PROJECT APPLICABILITY ON REAL-WORLD APPLICATIONS:-

1) This system can be used by students to monitor and improve their digital habits.
2) It can also be extended into mobile apps or web-based tools for wider use.

CONTRIBUTION / FINDINGS OF THE PROJECT:-

- Digital distraction directly affects productivity.
- Simple analysis can help improve habits.
- Users respond better to suggestions than raw data.

LIMITATION / CONSTRAINTS OF THE SYSTEM:-

1) Does not store user history
2) No real-time tracking
3) Uses simple rule-based logic only

CONCLUSION:-

- This project successfully demonstrates a simple AI-based system that analyzes digital usage and provides meaningful suggestions.
- It helps users make better decisions and improve their productivity using basic AI concepts.

FUTURE ENHANCEMENTS:-

- Add more detailed analysis.
- Use machine learning for better prediction.
- Store user history.
- Add graphical interface.

REFERENCES:-

- Python Documentation
- Online tutorials
- Class notes